The Journal of Supercomputing (2025) 81:737
https://doi.org/10.1007/s11227-025-07182-1

Intelligent device to device handover management techniques for 5G/6G and beyond

S M Topazal[1] · Shayla Islam[1] · Raenu A./L. Kolandaisamy[1] · Mohammad Kamrul Hasan[2] · Ahmad Fadzil Ismail[3] · Nur Hanis Sabrina Suhaimi[2] · Huda Saleh Abbas[4] · Muhammad Attique Khan[5] · Kamal Ali Alezabi[6]

Accepted: 13 March 2025

Abstract

As mobile networks grow to 5G, 6G, and beyond, uniform Handover Management (HOM) becomes important to ensure continuous connectivity and an improved user experience. This research investigates the complexities of HOM in next-generation (NZ) networks, concentrating on the problems and solutions for a smooth transition between various network types and standards. Device-to-Device (D2D) requires a fast, intelligent, and reliable handover (HO) decision system to deliver seamless mobility and ongoing service. With the exponential growth of Internet of Things (IoT) devices, D2D communication, augmented reality (AR), virtual reality (VR), and other technologies, efficiently managing the handover process which is critical to ensuring low latency, high data rates, and huge connectivity. The study investigates advanced methods such as machine learning (ML), long short-term memory (LSTM) model, and deep learning (DL) for improving handover management and reducing service disruptions. This paper is in accordance with sustainable development goals (SDGs), Goal 9: specifically, industry, innovation, and infrastructure, by linking the generations inside the network using creative HO techniques. Through improving digital infrastructure, encouraging innovation, and guaranteeing fair access to high-quality, reliable, and robust communication networks, the results seek to advance sustainable manufacturing. This in-depth examination of handover management for 5G, 6G, and beyond offers a thorough study that researchers, policymakers, and network operators working toward a sustainable digital future will find invaluable.

Keywords Handover (HO) · Fifth-generation (5G) · Sixth-generation (6G) · Internet of things (IoT) · Machine learning (ML) · Deep learning (DL) · Long short-term memory (LSTM) · Handover management (HOM) · Device-to-device (D2D)

Extended author information available on the last page of the article

Published online: 12 April 2025

Abbreviations

4G	Fourth-generation
5G	Fifth-generation
6G	Sixth-generation
B5G	Beyond fifth-generation
EMR	Ericsson mobility report
HD	High definition
D2D	Device-to-device
uRLLC	Ultra-reliable low latency communications
MTC	Massive machine-type communications
waves	Millimeter waves
ML	Machine learning
AI	Artificial intelligent
eMBB	Enhanced mobile broadband
SNR	Signal-to-noise ratio
NN	Neuron network
RSSI	Received signal strength indicator
MIH	Media independent handover
FPMIP	Fast proxy mobile IPv6
LIM2	Learning-based intelligent mobility management
SIR	Signal to interference plus noise ratio
NS3	Network simulator 3
SDN	Software-defined networks
ECDF	Empirical cumulative distribution function
SVM	Support vector machine
HO	Handover
HOM	Handover management
UAV	Unmanned aerial vehicle
DRL	Deep reinforcement learning
mUAV	Multi-unmanned aerial vehicle
NR	New radio
MT	Mobile terminal
MTM	Mobile terminal mobility
BS	Base station
RSS	Received signal strength
HSR	High-speed rail
UHM	Ultra-high-mobility
ECDF	Empirical cumulative distribution function
RL	Reinforcement learning
EPC	Evolved packet core
V2V	Vehicle to vehicle
UMS	User mobility and speed
V2X	Vehicle to everything
IoT	Internet of things
gNB	Next generation node B
V2I	Vehicle to infrastructure

LTE	Long term evolution
HetNets	Heterogeneous networks
UE	User equipment
UoC	Quality of service
LSTM	Long short-term memory
GSO	Grid search optimization
M2M	Machine-to-machine
3GPP	3Rd generation partnership project
LENA	Low energy neutrino astronomy
RSRP	Reference signal received power
SINR	Signal-to-interference-plus-noise ratio
EPC	Evolved packet core
eNB	Evolved node B
TCP	Transmission control protocol
TTT	Time-to-trigger
USE	User equipment
CDF	Cumulative distribution function
MSE	Mean squared error
DL	Deep learning
UDN	Ultra-dense network
S-BS	Serving base station
MMS	Mobile movement speed
HHO	Horizontal handover
VHO	Vertical handover
MCHO	Mobile-controlled handoffs
MAHO	Mobile-assisted handoff
NCHO	Network-controlled handoff
HOCP	Handover control parameter
RAT	Radio access technology
uLL	Ultra-Low Latency
MRA	Multi-radio access
multi-RAT	Multi-radio access technology
MSE	Mean squared error
AP	Access point
TCP	Transmission control protocol
SO	Signaling overhead
SA	Security and authentication

1 Introduction

Recently, wireless communication networks have witnessed an unprecedented increase in traffic and data rate demands as more wireless devices are getting connected. The Ericsson mobility report (EMR) highlights an increase in network traffic by over 56% (percent) only in the first quarter of 2020 [1], bringing network traffic management to be one of the primary issues for 5G networks and beyond [2]. A significant change in technology and infrastructure is required as telecommunication networks progress towards the 5G/6G of mobile communications. Additional application use cases, with many users and D2D communication with different performance needs, may all be supported by 5G/6G and beyond. D2D communication refers to the process of communicating directly between objects, without requiring any infrastructure node to facilitate data transfer [2]. Within the context of global digital transformation, collaboration and communication between devices through device-to-device communications are essential components [4]. One of the most significant technological innovations that make the Internet of Things (IoT) possible is this. As a result, it is predicted that it will be a vital part of the Internet of Things [2–4]. In the studies investigated in 2021, the number of IoT-connected devices of approximately 13.8 billion, whereas non-IoT-connected devices were nearly 10 billion. The advent of modern applications in the form of virtual reality, high-definition (HD) video streaming, etc., is further exacerbating the issue [3, 4]. It has been highlighted in [4] that HD video streaming contributes to more than half the data traffic and is increasing alarmingly. Resultantly, the demand for making a rapid shift towards 5G/6G networks is imminent [1], where the capacity of the networks is expected to increase manifolds [5–7]. Mechanisms like massive machine-type communications (mMTC) and ultra-reliable low-latency communications (uRLLC) will help offer a higher degree of bandwidth to meet the desired bandwidth demands [8]. Apart from the enhanced application spectrum, the enhanced technology spectrum is raising challenges in mobility management. The use of the Internet of Things (IoT), for example, is becoming ubiquitous in various applications like agriculture [9], healthcare [10], smart homes [3] and cities [11–13], etc. Similarly, the concept of autonomous vehicles and the development of modern architectures like Vehicle to Everything (V2X) (that comprise Vehicle to Infrastructure (V2I), Vehicle to Vehicle (V2V) etc.) [14, 15] has increased the data transmission demands at immensely higher rates that cannot be catered to by the existing 4G and long-term evolution (LTE) networks [3].

The advent of 5G/6G networks is a significant achievement to overcome some of the challenges presented above [12]. These networks possess higher bandwidth capacity by having more comprehensive resources and infrastructure, thus revolutionizing many sectors like healthcare, transportation, etc., through reduced latency rates and giving rise to new applications like augmented reality [13]. Additionally, 5G/6G entails energy-efficient mechanisms, improved network slicing capabilities, and connects many devices through Machine-to-Machine (M2M) [16] and IoT communication. Such improvements facilitate a transition towards a digital world, where automation can be introduced in many processes. Despite the benefits that 5G/6G offers, there are some associated challenges due to the particular communication framework [2, 6]. The complex architecture possesses unique challenges requiring innovative and software-led solutions. One such issue is handover management [7, 17]. Handover is referred to as seamlessly switching between the base stations (BSs) to ensure undisrupted communication. The fundamental cause of such HO is the mobility of the users as they move from the coverage area of one BS to another at high or mode-rate speeds [17]. The handover process gives to formidable challenges caused by numerous factors, as discussed below:

- The 5G/6G utilizes higher frequency bands to attain higher data transmission rates. Such higher frequencies are more prone to attenuation and short-range traversing. Thus, a need for rapid handover possessing higher complexity levels surfaces. Rapid and frequent switching between Radio Access (RA) technologies is increasingly taking place.
- The 5G/6G networks are associated with dense Heterogeneous Networks (HetNets). These networks comprise a diverse set of small and macro-cells along with other access points. This architecture intensifies the HOM problem. Advanced mechanisms are required to ensure seamless coordination and handover across various cells.
- To support real-time applications and services (AR, telemedicine, etc.) requiring ultra-low latency rates, HOs serve as bottlenecks and fundamental units to ensure a smooth communication experience for users.

Thus, addressing the HOM in 5G/6G networks plays a crucial role in unleashing the full potential of such communication networks [18]. Handover management is the process of moving an ongoing connection from one BS or network cell to another when users travel through the network coverage area. This technique is essential and directly influences the Quality of Service (QoS) and Quality of Experience (QoE) for consumers. Apart from ensuring seamless communication, such mechanisms ensure that resources are utilized optimally and QoS thresholds are maintained. Offering efficient HOM processes, 5G networks can help deliver unparalleled user experience [19] and ensure robust connectivity demands. The handover methods employed in 5G networks have significantly improved over previous generations. This is mostly due to the implementation of cutting-edge technology like edge computing, network slicing, and beamforming. However, the predicted 6G networks introduce new dimensions to network management due to variables like increased user mobility, a broader range of service requirements, and the incorporation of cutting-edge technology such as quantum computing, artificial intelligence (AI), and ML. To ensure that 5G, 6G, and future network systems work seamlessly, HO management strategies must be reassessed and improved. Numerous techniques have been explored and presented to improve the HOM process in 5G/6G networks and beyond [20]. In classifying such techniques, articles focus on three types of discussions: conventional methods, machine learning approaches, and survey articles to review the existing techniques. The conventional techniques for HO involve approaches like signal strength-based HO, hysteresis-based HO, load-based HO, etc. Machine learning algorithms are extensive and involve methods like shallow learning, deep learning, etc. The survey articles generally target one of the domains (machine learning, etc.) and present some of the existing research available. This article mainly focuses on deep learning-based methods used in modernizing HOM. These survey studies, along with deep learning techniques such as LSTM, reinforcement learning (RL), and others, have been investigated as part of the literature study.

1.1 Paper contribution

In the present research, we explore the handover management scheme in 5G/6G network architecture in a wide sense, delving into aspects such as network traffic, signal degradation, network congestion, and quality of service based on the model that has been presented. Here is a brief overview of this paper's contribution below:

- The technical challenges of HO delays will require more in-depth evaluations as 5G/6G network strategies advance.
- It presents the technical factors, advantages, and disadvantages that may be used to improve the HO latency using an LSTM model.
- This paper presents an overview of the LSTM model.
- An analysis of how upcoming technologies affect the handover delay in 5G/6G networks.
- The comparative analysis is responsible for identifying and addressing various open research issues and future directions that have improved outcomes, such as reducing the amount of time required for downloads and the number of handover decision-makers, among other things.

1.2 The organization of paper

This paper provides in-depth research and discusses funding approaches for handover management in 5G/6G networks based on ML. The first section gives the necessary knowledge of HO management to comprehend the idea. Section II presents some existing articles (survey articles and machine learning-based handover management). Two methods use LSTM for handover management, highlighting their similarities and differences, and the adopted approach is presented in Sect. 3. Section 4 presents some of the key results of these two articles with their proposed model. The study concludes with some of the prospects for future research discussed in Sect. 5. Figure 1 represents the paper structure.

1.3 Related works

Several studies on 5G/6G wireless communication mobility management have been published and referenced in Table 1 below (summary of the recent research articles on handover management). The research discusses important aspects of mobility, such as handover, cell selection, cell reselection, and the difficulties of cellular communication caused by differences in transmitted power and cell coverage area in deploying heterogeneous networks (HetNets) [21]. In addition, the authors provide techniques and solutions to address these difficulties. In [19], the IoT and Uncrewed Aerial Vehicle (UAV) networks are examples of contemporary cellular networks that make use of Deep Reinforcement Learning (DRL). By the utilization of DRL, the details of the process by which network entities (User Equipment, Drone etc.) learn the best policies for things like HO and offloading

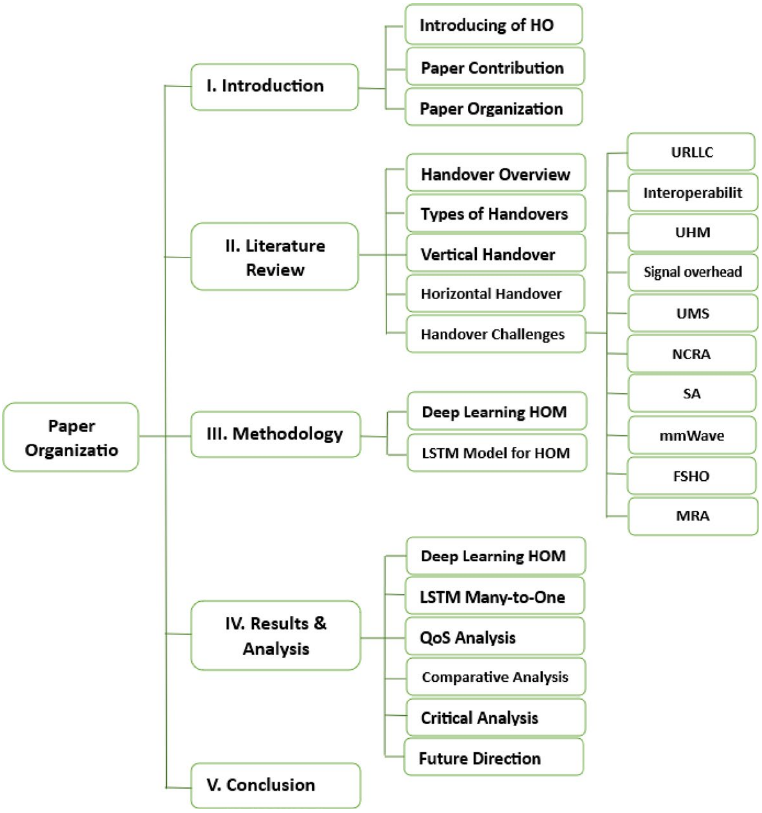


Fig. 1 The paper organization

decisions, cell and channel reselection and selection, a dependable connection of multi-UAV networks, and mobility patterns. In addition, the authors explore areas such as measurements for HO performance, different kinds of HO failure, how New Radio (NR) differs from LTE, mobility enhancers, and possible research obstacles and remedies for HOM [19].

The network can maintain active connections during mobile terminal (MT) mobility due to HO management. Additionally, it is possible to distribute the load of the network equitably across a variety of locations. Three steps may be distinguished within the HO process: the start stage, the decision stage, and the execution stage [22]. Depending on certain circumstances, such as the radio bearer's deterioration or the network's congestion, the HO initiation is the component that is accountable for initiating the HO. The determination of the new Access Point (AP) that is the most suitable is made during the stage of HO decision-making. Before a final selection is made, this stage involves the consideration of several characteristics, such as the signal strength of neighboring APs, the available radio resources, and so on. Finally, the necessary signaling exchange is carried out at the last stage to reestablish contact and reroute data along the new channel [22].

Table 1 Summary of the recent research articles on handover management

Article	Objective	Methods used	Type	Outcomes
[18]	Survey of intelligent HO management in 5G networks	Review, ML algorithms training	Survey	Identification of challenges and ML applications in 5G HO management, future research directions
[19]	Utilizing artificial intelligence and machine learning to support 5G technologies	–	Survey	Overview of AI and ML Applications in 5G Technologies
[20]	Analyzing handover performance in 5G small cells	–	Survey	Examination of handover performance in 5G small cells
[21]	Optimal handover mechanism for 5G vehicular networks	Reinforcement Learning, Kalman filter	Simulation	Efficient handover management in 5G vehicular networks
[19]	Predictive handover using LSTM and SVM	LSTM, SVM	Simulation	High prediction accuracy and reduced handover latency for mobile devices
[22]	Online learning-based mobility management in 5G	Online learning, SARSA, Kalman filter	Simulation	Improved handover operation in 5G and beyond networks
[23]	Software Define Network and machine learning-based handover management	–	Survey	Suggested solutions for handover management in 5G HetNets
[24]	Hybrid handover technique with LSTM and SVM	LSTM, SVM	Simulation	High prediction accuracy and a significant reduction in handover latency
[25]	Robust handover mechanism with LSTM and SDN	LSTM, SDN, machine learning	Simulation	Improved handover mechanism with reduced hand-over failures and number of HOs
[26]	LSTM network for Radio Resource Management	LSTM, Grid Search Optimization (GSO)	Simulation	Improved indoor guaranteed rate compared to existing methods
[27]	Handover optimization in emerging 5G radio networks with beamforming-capable nodes. DL is utilized for dynamic policy optimization based on user behavior and channel responses	Feature extractors, dense layers, supervised training based on 3GPP model, and optimization to reduce handovers	Simulation	Significant reduction in handovers without compromising system throughput, proactive reaction to signal changes

Table 1 (continued)

Article	Objective	Methods used	Type	Outcomes
[28]	The scheme of HO management for Next Generation Networks	Multi-layer many-to-one LSTM architecture, multilayer LSTM AutoEncoder with MLP	Simulation	An improved number of users finalizing downloads by 18% reduced download time compared to the benchmark scheme
[29]	The use of ML models to predict future blockages and UE-based link status monitoring are two conventional methods for guaranteeing uninterrupted connection	Deep learning, Light detection, and ranging (LiDAR)	Simulation	Over 18% improved expected accuracy can be achieved by enabling proactive HO that ensures an accurate communication connection
[30]	Evaluate the performance of LSTM in different strengths and evaluate the accuracy and speed of SP-LSTM forecasts	LSTM, SP-LSTM (Speed-Optimized LSTM)	Survey	The effectiveness of LSTM in identifying long-term relationships in sequential data is contrasted with SP-LSTM's focus on providing quick and accurate predictions
[68]	Survey of D2D communication in 5G/6G networks	Review of D2D applications, classifications, and challenges	Survey	Identifies key applications, classifications, and challenges in D2D communication. Highlights the potential of direct device interaction in enhancing network efficiency
[69]	Dynamic D2D communication using AI in 5G/6G networks	Distributed AI framework for network topology management	Simulation	Demonstrates AI-based approaches for optimizing D2D communication in dynamic environments
[70]	UAV-assisted D2D communication in future networks	A simulation-based study on UAV-enabled connectivity	Simulation	Shows how UAVs improve D2D efficiency, reduce latency, and enhance coverage in large-scale networks

HetNets are designed to expand the network's coverage area and the number of users it can accommodate. The small cells may be introduced to the network in the presence or absence of macro-cells, in open or covered locations, and in dense or sparse configurations. All of these factors could be controlled by the network [23]. The ML techniques are the primary distinction between the existing deployment of 4G/5G networks and beyond-fifth generation (B5G)/beyond-sixth generation (B6G) networks currently being developed. The studies [24] strongly emphasized the decrease of the blocking error of HO at the cell border of the BSs. A flexible HO method was proposed by [24] to reduce the time HO delays take in considerably congested cellular networks. Within the framework of this strategy, the HO procedure was bypassed by many BSs distributed over the users' trajectories. This scheme execution for the single handover skipping has major benefits in various practical phenomena, and the evaluation and analysis of this scheme have those advantages [25].

As a result, 5G/6G mobile technology has been used to increase system capacity and provide connections everywhere [31]. Therefore, millimeter wave (mmWave) bands have been recommended as the most suited and effective option for 5G/6G and beyond mobile networks [26–28]. This is because all aspects of mobile networks need to go through a paradigm shift because of the larger bandwidth that today's consumers create. The Ultra-Dense Network (UDN) is one of the most important ideas behind the racecourse. Networks with more cells than active users are called UDNs [31, 32]. The HO performance is a characteristic frequently evaluated in cellular networks because it is an excellent indication for exhibiting effective mobility strategies. The number of drones required to deliver network access services to certain terrestrial customers may vary depending on the capabilities of the drones that are part of the network. Cellular networks make use of a wide range of HO decision-making algorithms, including the Serving Base Station (S-BS), Received Signal Strength Indicator (RSSI), Mobile Movement Speed (MMS), distance between the BS and UE, limited capacity of BSs, Signal-to-Interference-Plus-Noise Ratio (SINR), and cost functions, weight functions, control of fuzzy logic, and ML with DL technology, respectively [33, 34].

Within the framework of the technique provided in [34] the HO procedure can be bypassed by several BSs distributed along the users' paths. This paper assesses and examines the benefits of using a single HO skipping strategy in various real settings. The study demonstrates that single handover skipping systems provide significant benefits, such as reduced HO frequency, lower signaling overhead, and increased user satisfaction by ensuring persistent connectivity. The study finds that implementing this technique in real-world circumstances could result in more efficient network management and resource use.

2 Literature review

According to Mollel et al. [18], presented a survey that discusses the challenges posed by handover management in the 5G/6G and cellular networks. The discussions are specific to the novel enabling technologies in mmWaves and the IoT.

Machine learning models have been presented to address these challenges. Such techniques have been reviewed under the modern taxonomy of the data sources [19]. Analyzing numerous ML and AI techniques for HO management in 5G/6G communication networks [35]. Performance requirements in mMTC, eMBB, and uRLLC have been incorporated. Mobility management in such networks has been extensively discussed using reinforcement learning techniques. In [19, 20] analyzed the small cell networks in the context of 5G/6G and beyond for the HO performance analysis. A comparison of HO in 4G, 5G, and 6G is also covered to comprehend further technological advancement and how it affects HO solutions [35]. The smaller coverage areas with higher BS frequencies have been considered. For practical considerations, urban communication channel modeling has been simulated. Signal-to-Noise Ratio (SNR) and Received Signal Strength Indicator (RSSI) have been used to analyze the handover performance analysis and implications under such communication conditions. Kosmopoulos et al. [36] present optimal HO mechanisms in the 5G/6G communication networks. A novel technique called Media-Independent Handover (MIHO) and Fast Proxy Mobile IPv6 (FPMIP) standard has been considered. The HOM has been presented in reactive and predictive scenarios considering the velocity of the users and other network conditions. The study has been found to offer promising results compared to the earlier versions of FPMIPv6.

In [37] offers a novel neural network-based approach towards HO management in the 5G/6G cellular networks. By incorporating the mobility data, the use of neural networks helps identify the optimal HO paths for the groups of users. This helps attain efficient network performance. It has been found that the developed networks attain over 90 percent of the accuracy. According to Karmakar [38] presents an online learning mechanism for mobility management in 5G networks called Learning-based Intelligent Mobility Management (LIM2). The method adopts a Time-to-Trigger (TTT) mechanism and hysteresis values to allow mobility management in high-speed scenarios. Kalman filtering was used to predict SNR. On top of that, the method employs a reinforcement learning approach for making HO decisions. The experimentation has been performed in Network Simulator 3 (NS-3) to determine the HO performance in high-speed mobility scenarios. In [39] presents the scenario of HO management in heterogeneous networks (HetNets). Using mmWave in 5G/6G communication has employed a high band spectrum. Both short and long-range cells have been considered to analyze the performance. The complexity of the networks has been considered for performance implications. Numerous HO management techniques have been presented in this context that focus on using software-defined networks (SDN) and machine learning.

Kaur et al. [40] presented hybrid HO management in 5G networks by using a DL model, LSTM, and Support Vector Machine (SVM). The models have been developed using predictive analysis that involves parameters like speed, location, etc. The method has been found to attain over 99% (percent) predictive accuracy and helps attain reduced latency in HO management. In [41] investigates, a robust HO mechanism using LSTM and SDN. The HO performance has been studied using numerous ML algorithms. LSTM has been found to outperform all such models by demonstrating an R2 score of 0.998. Additionally, the failure rate of the HO management has been reduced by 30 percent. According to Balmuri et al. [42] present an LSTM model for radio resource management in 5G communication networks. A grid search optimization (GSO) algorithm has been adopted, and the power and bandwidth of the users have been considered. The undesired resource allocation has been curtailed, and frequency inter-leaving and guard insertion have been employed to reduce transmission losses. Improvement in various parameters has been found. In [43] presents, a high-level handover management model in 5G/6G communication systems. A deep learning model has been developed for making HO decisions in varying conditions and real-world impairments. Two stages of training have been involved, and the overall handover scheme has been optimized. Without compromising on the throughput, the approach optimizes the HO decisions. According to Ali et al. [44] investigated a HO management scheme in the context of next-generation self-organized networks. By leveraging data traversing along the protocol stacks, an intelligent HO management model has been developed using multilayer many-to-one LSTM. The number of users able to complete their downloads during HO scenarios has increased by 18% (percent) compared to a benchmark handover scheme.

Device-to-Device (D2D) communication in 5G and 6G networks enables direct interaction between devices without routing data through centralized network infrastructure. This approach significantly enhances data transmission efficiency, reduces latency, and improves overall network performance. Several recent studies have explored various aspects of D2D communication in NZ networks. Alam et al. [68] conducted a comprehensive survey on D2D communication within 5G and beyond (5 GB) networks, analyzing its applications, classifications, and inherent challenges. Liu et al. [69] investigated dynamic D2D communication in 5G/6G networks, emphasizing the role of distributed AI in managing evolving network topology due to device mobility. Their study highlighted AI-driven adaptability in real-time environments. Furthermore, Zhang et al. [70] examined UAV-assisted D2D communication, demonstrating how unmanned aerial vehicles enhance efficiency and reduce latency in future wireless networks. In comparison to these studies, the current research integrates advanced AI models, specifically LSTM, to optimize HOM and positioning estimation in D2D communication. This approach provides a more adaptive and intelligent HO decision-making process, ensuring seamless connectivity, while addressing network congestion and signal degradation issues. By leveraging LSTM's ability to process sequential data and predict mobility patterns, the analyzed model enhances HO accuracy and reduces service disruptions, ultimately improving overall network performance. A summary of these articles has been shown in Table 1.

2.1 Types of handovers

Handover may also be divided into several categories, each determined by the technology of the providing network and the destination network. The horizontal handover (HHO) and the vertical handover (VHO) are the two main types of HOs [45, 46]. The network interface does not change during the HHO, and the access points continue to utilize the same technology [19]. In the process of VHO, the access technologies are distinct from one another, and several network interfaces are used.

For example, the user transitions from the terrestrial cellular network to the satellite technology, as shown in below Fig. 2. Also, there are three methods used by cellular networks to execute handoffs [45]:

- Mobile-Controlled Handoff (MCHO)
- Mobile-Assisted Handoff (MAHO)
- Network-controlled handoff (NCHO) determines these methods using the network management system.

In addition, the performance matrices and simulation scenarios utilized in evaluating the system are discussed [46]. A simulated environment was constructed in MATLAB to evaluate the efficiency of Handover Control Parameters (HOCP) in the HO performance of the B5G network. As shown in below Fig. 3, 61 gNBs have been deployed in a simulated environment, and the distance coverage of each BS is 220 m. The user equipment within the simulation environment is configured to travel in a straight line but randomly within the green limited circle with eight directions (45 degrees). They are expected to go through BSs at 25, 45, 85, 122, 165, and 220 km/h, the six possible scenario speeds. When a user entity hits the boundary of the green region, it will randomly shift the direction in which it is moving [47].

2.2 Vertical handover (VHO)

The handover decision process and execution procedure are two distinct phases in heterogeneous wireless networks. The timing of the HO is determined by the mobile node and network simultaneously in the decision process. Three primary processes characterize the HO delay process: 1. Discovery phase, 2. Resolve the configuration time, and 3. Time required to register a network [48]. Several subtle differences

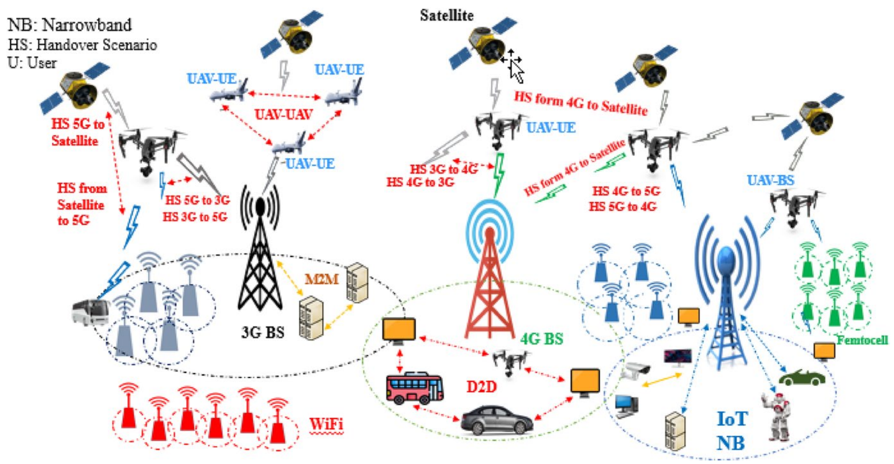


Fig. 2 The scenarios of handover with associated drones in the future and upcoming mobile networks [45]

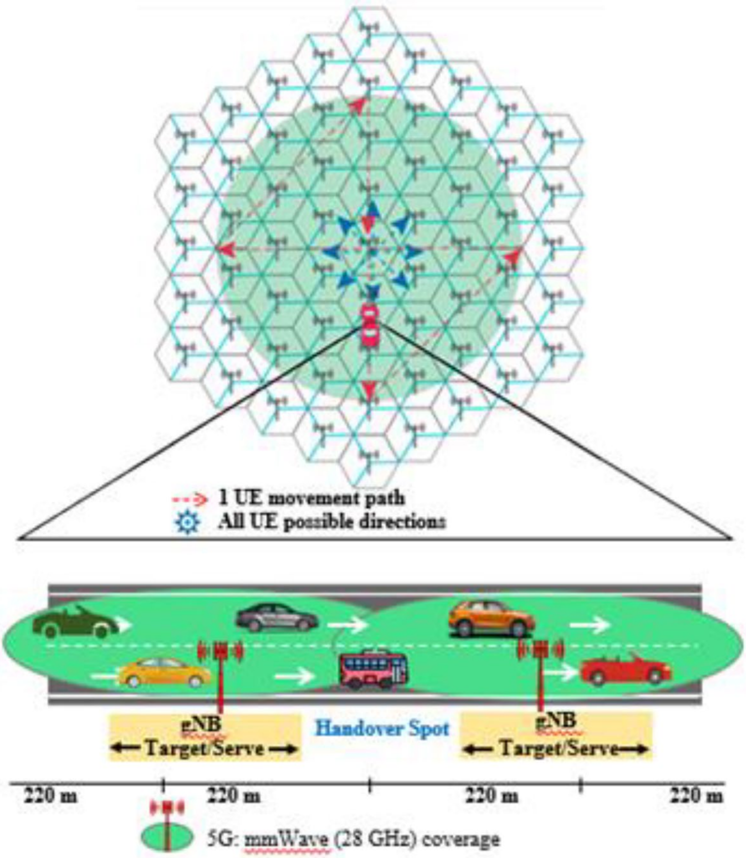


Fig. 3 The scenario for the implementation of the B5G network [47]

exist between the typical HHO process and the VHO mechanism. They are further divided into two distinct categories based on the signal intensity. Upward VHO and downward VHO are the starting points for the first categorization. When a network with limited coverage and a high data rate transitions to a network with more coverage and a lower data rate, this occurrence is known as an upward vertical handover [45, 48, 49].

2.3 Horizontal handover (HHO)

The horizontal HO is characterized by the fact that it always occurs inside the same radio access technology (RAT), which implies that the HO process occurs within various cells of the same network [48]. When the serving access router in a homogenous network becomes unavailable because of MT's mobility, HHO is often mandated as a standard practice [49]. VHO may be required in heterogeneous networks for convenience rather than connection. This is because of the nature of the networks. For the most part, HHO makes the decision regarding the handoff based on the received signal strength (RSS) [50, 51].

2.4 Handover challenges

Handover management (HOM) in 5G/6G wireless networks is becoming increasingly difficult as new technologies are developed specifically for B5G/B6G systems to fulfill their needs. We discussed the most notable HO optimization difficulties in B5G/B6G networks by applying new technologies to support the network. These challenges involved the introduction of new technologies [52].

2.4.1 Ultra-Low Latency Communication (uRLLC)

Mobility execution is one of the most important parts of wireless interactions and is also relevant to uRLLC. Latency, reliability, mobility, and stability are key components of uRLLC services [52–56]. A low-latency connection is required for several services and applications, including 5G/6G activation, which requires fast and seamless HO methods with no delay.

2.4.2 Interoperability

It refers to the capacity of different networks, access methods, devices, or systems to communicate with one another and perform harmonically, while working together [57]. Additionally, in B5G/B6G, users' mobile devices will be linked to several mobile wireless networks and access technologies to guarantee connection and availability. This would allow users to travel and switch between different networks [52] effortlessly [57].

2.4.3 Ultra-High Mobility (UHM)

In high-speed mobility situations, like high-speed rail (HSR) systems, guaranteeing reliable broadband wireless connection is a big challenge for developing B5G/B6G systems. Additionally, B5G systems, like 6G, are anticipated to sustain mobility speeds of up to 1000 km/h. See [52] for a presentation of the key issues with HO in highly mobile systems. Many users experience a greater HO rate when they are highly mobile. The system could not have enough time to finish the HO operation because UEs can transit over the overlapping coverage region of the two BSs fast mmWave [52, 58].

2.4.4 Signaling Overhead (SO)

The increased complexity of handover procedures in 5G/6G networks leads to excessive signaling overhead. Traditional HO mechanisms involve multiple steps, including measurement reporting, decision-making, authentication, and reconfiguration, contributing to delays. Future networks must adopt AI-based predictive HO models to minimize redundant signaling, while maintaining accuracy [45–47, 71].

2.4.5 User Mobility and Speed (UMS)

Extreme mobility scenarios such as High-Speed Rail, unmanned aerial vehicles (UAVs), and satellite-based networks challenge conventional handover mechanisms. High-speed users spend less time within a cell, leading to frequent and unnecessary HOs (Ping-Pong effect), increasing latency and packet loss. AI-driven mobility prediction models and multi-connectivity strategies are being developed to counteract these issues [46, 47].

2.4.6 Network Congestion and Resource Allocation

Densely deployed 5G/6G small cells cause overlapping coverage areas, leading to increased network congestion and resource contention. Load balancing and interference management strategies must be optimized for efficient HO execution. Edge computing and distributed AI models are being explored to improve resource allocation dynamically [72, 73].

2.4.7 Security and Authentication

With the adoption of decentralized architectures and blockchain technology in 6G networks, securing handover authentication without adding excessive delay is a challenge. Blockchain-based handover authentication schemes are being investigated to enhance security, while optimizing computational overhead and ensuring seamless mobility [58, 59].

2.4.8 Millimeter Wave (mmWave)

A variety of mmWave bands are utilized by 5G/6G mobile networks to increase the interaction volume significantly. Regarding directivity, sensitivity to obstructions, and large propagation loss, mmWave interactions differ considerably from other existing interaction systems. Full use of these mmWave interface characteristics raises several issues [59]. Given this context, [60] provided a comprehensive overview of the recently proposed 5G/6G mmWave wireless system concepts as well as a selection of the most important mmWave radio propagation models that have been produced worldwide, thus far [52, 59, 60].

2.4.9 Fast and Seamless Handover

Moving the user equipment from one BS to another with as little disturbance as possible to the continuing communication is referred to as a fast HO. A seamless HO is described as a HO procedure that, when a HO happens, does not disrupt the connection between the user equipment and the network and the data linked to the network [61]. In addition, the purpose of seamless HO is to guarantee that the transfer of data from beginning to end is made without interruption if there is a disruption in the connection [62]. Fast and seamless HO is a significant difficulty in the B5G system. This is because B5G demands ULL and 100% connectivity to serve various service instances, which require ULL and an uninterrupted connection [52, 63, 64].

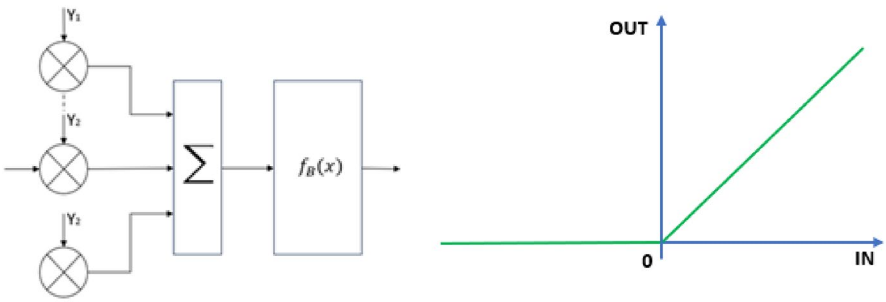
2.4.10 Multi-Radio Access

Using multi-radio access technology in B5G/B6G networks can improve traffic capacity, spectrum efficiency, connection density, and coverage [65]. The capability of the user equipment to utilize numerous wireless communication technologies concurrently is called multi-RAT. Disruption, latency, and complexity are just a few of the HO management issues that worsen when networks are diverse. The multi-RAT network situation poses additional difficulties for HO management due to the potential for increased frequency interference between networks, which in turn might impact signal quality. The difficulties of HOM in B5G/B6G multi-RAT networks are exacerbated by delay, among other factors. A major obstacle in managing handover in multi-RATs is the intricacy of the process [52, 65].

3 Methodology

The system models for two LSTM-based handover management shown in [43, 44] have been analyzed. The selection of these specific papers within the review paper is guided by their exemplification of distinct yet highly effective approaches to address the complex challenges of HO management. These papers were strategically chosen to demonstrate the practicality and impact of modern machine learning techniques, specifically DL, and LSTM, in HOM. In [43], it focused on a DL model that integrated positioning estimation and HO management, showcasing its ability to improve HO decisions and enhance network performance. Meanwhile, Ali et al. [44] delved into developing a Many-to-One LSTM model for HO management, offering insights into its real-world implications and its capability to optimize HO scenarios. By including these papers in the review, the aim is to provide readers with a tangible understanding of how advanced ML models can address key performance factors and contribute to a seamless communication experience.

As shown in below Fig. 4a, a neuron is the fundamental component that builds Neural Networks (NN). Several additional neurons are interconnected to each neuron, and each neuron performs two fundamental actions throughout each entrance. Activation is the second operation performed after the first one, which is a weighted sum of the outputs of prior neurons connected to the present one, is passed through. In neurons, the behavior of a neuron is defined by its activation function. The function must be simple to guarantee a speedy operation, as well as to extract and transmit essential information. The logistic or sigmoid function, which is derived from the logistic regression model, is the most commonly used activation function. As a result of the vanishing gradient issue and the fact that it requires a significant number of computational resources, this function is not frequently utilized in deep neural networks (DNN).



(a) (b)

Fig. 4 Single/individual neuron 4 (**a**), rectified linear unit (ReLu) function 4 (**b**) [43]

$$x = \max(0, y)[43] \tag{1}$$

Following Eq. (1), the rectified linear unit (ReLu) is the activation function that is utilized the most frequently and yields the best results. This function is shown in below Fig. 4b. Since the derivation of the function is constant and requires inexpensive computational effort, ReLu does not experience the problem of vanishing gradient. In the last layers of neural networks (NN), other activation functions, such as linear, step function, or softmax, are utilized almost entirely. These functions are responsible for determining the shape and size of the output.

3.1 Deep learning handover management

The 3rd generation partnership project (3GPP) model presented in [66] has been employed as part of the communication scenario in this study. It uses a reactive HO approach in LTE networks [4]. A handover occurs to a candidate beam if the reference signal received power (RSRP) value exceeds a selected threshold. The data from this has been used to train the model. Some practical considerations in the data involve issues like ping-pong HOs [67], fast channel shadowing, and channel noise. The DL model developed includes two physically separated components, i.e., a positioning estimation model and the HO management model. The position estimation is used for feature extraction, where 224-dimensional vectors are converted into 2D coordinates of the user positions. Thus, positional data, the serving beam information, and the historical RSRP information are handed over to the HO management model. Three stages of training are involved:

First, the positioning model has been presented which was followed the 3GPP guidelines. The LTE model as accurate labels, and finally, optimization of HOM using future dataset samples. This has been shown in Table 2. The model developed as part of HO management has been shown in below Fig. 5. In Eq. (2), we can see the function that predicts the actual label, where is a decreasing vector of 19 weights and is a matrix of 19×226 including both the current and 18 subsequent RSRP data.

Table 2 Summary of the model stages [43]

Stage	Model	Input data	Output data
1	Positioning data	RSRP data	Current position
2	Handover blank	Current and past RSRP data Positioning data	Next SB index
3	Handover pre-trained	Current and past RSRP data Positioning pre-diction	SB Index

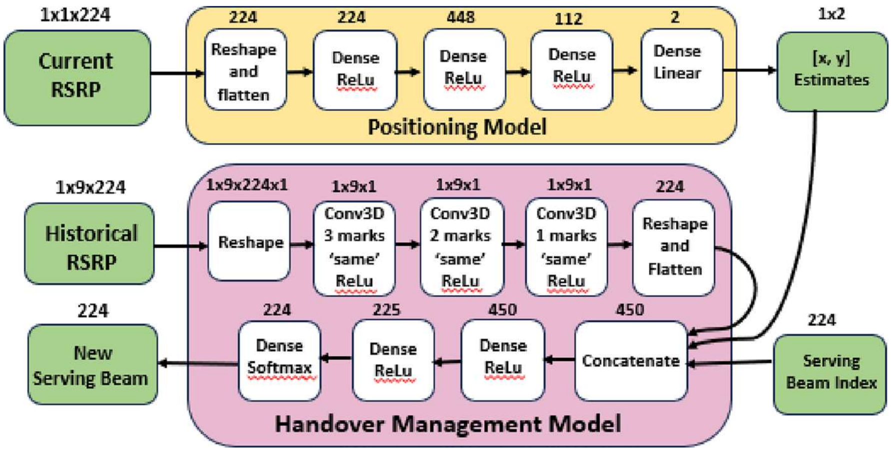


Fig. 5 Handover management (HOM) model [43]

$$\overline{x_{true}} = \text{arymax}\ (\overline{y} \cdot \overline{\text{RSRP}})[43] \tag{2}$$

Maximizing the weighted sums of the received signal strengths in successive measurements may determine the real beam index. Instead of using a single "true" label, the model is forced to choose just the high RSRP beams using the second function for true label prediction that is implemented in the second and third stages of training. As the most powerful beam accessible, this method returns weighted values for beams with 95% or higher RSRP.

3.2 Many-to-one LSTM model for handover management

A many-to-one LSTM model for HO management has been developed in the system model. The simulation scenario was initially created using the ns-3 low energy neutrino astronomy (LENA) LTE-EPC simulator. The communication scenario consists of an outdoor macro cell having three sectorial eNBs. UE positions are selected randomly at fixed distances, making reliable communication through TCP protocol across downlink and uplink. After the start of communication, mobility in UEs has been induced using a predefined pattern. Obstacles have been introduced to mimic a real-world scenario as coverage holes, noise, etc. The simulation incorporates a system bandwidth of 5 MHz. The mobility model considers a speed of 10 m/s covering a total distance of 4000 m. 20 potential runs for simulations have been introduced for making HO to the neighbors. The dataset in training and testing the LSTM handover model involves the data from the LTE protocol stack formulating a multivariate feature set. This involves the throughput, average number of packets received, cell IDs, communication rate, SINR, etc. The data was analyzed by leveraging the time dependencies contained in the data using the LSTM recurrent model. The model is shown in Fig. 6, where a multilayer many-to-one LSTM model has been developed. The model takes 200 steps, each having 84 features at the input. The data is passed in multiple batches [44, 46]. This section presents the key results generated by adopting the system models in the previous sections.

4 Results and analysis

This section presents the key results generated by adopting the system models in the previous sections.

4.1 Deep learning handover management

The positioning model developed in this study is accurate up to a few meters. The Euclidean error is 1.54 m compared to the true coordination on the targeted dataset. The median accuracy is 1.43 m on the unseen dataset. The cumulative distribution function (CDF) of the positioning prediction error with detailed median values has been shown below in Fig. 7. The developed HOM models have been compared to the 3GPP model and perform better. Even in the presence of real-world implications and uncertainty, the model's accuracy is found to be at par and follows an improvement trend. It has been found that a 4.2 percent reduction in HO count is observed

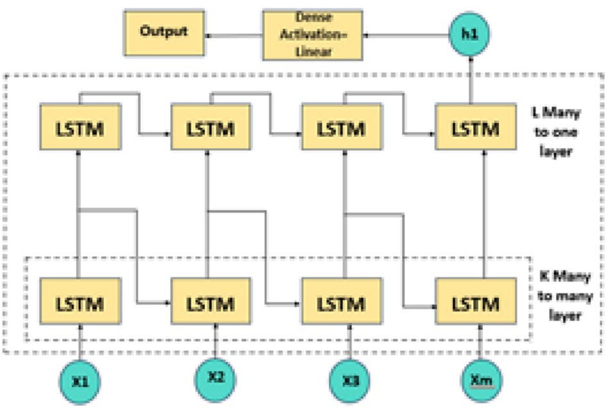


Fig. 6 LSTM many-to-one architecture [44]

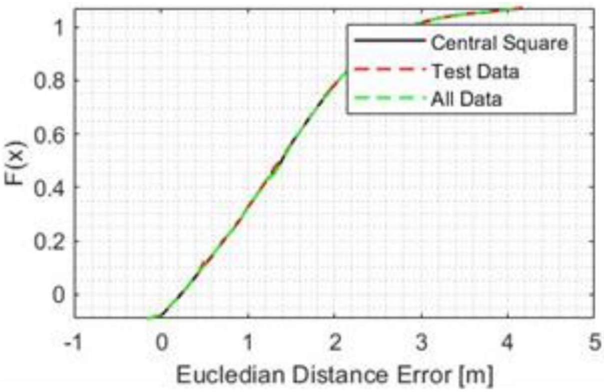


Fig. 7 CDF of positioning prediction error

with an uncertainty of 1, 55.8 percent reduction at the magnitude of 3, and 69.5 percent at the magnitude of 5. It has been found that the resulting predicted RSRP distributions for deep learning and 3GPP are identical. Some minor differences have been found along the tail distribution. The deep learning model demonstrated characteristics of the universal approximation model that imitate the detection function at close approximation levels [43]. This has been shown in Fig. 8. The average of 3GPP and deep learning outcomes has been taken, while generating the results again using MATLAB.

4.2 LSTM many-to-one architecture

The performance analysis of the LSTM architecture developed in the present scenario with hyperparameters chosen based on the lowest mean squared error (MSE) values

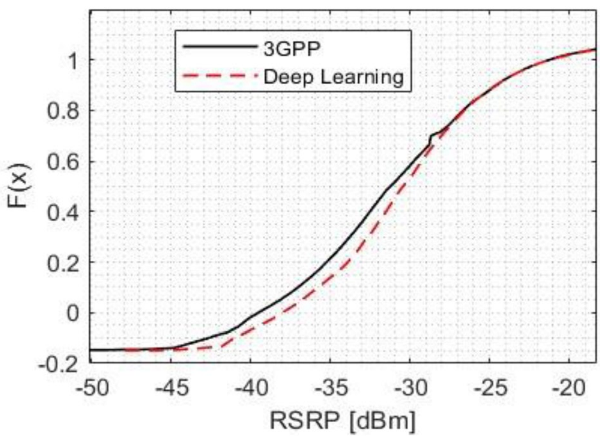


Fig. 8 The CDF of RSRP distribution against varying levels of uncertainty

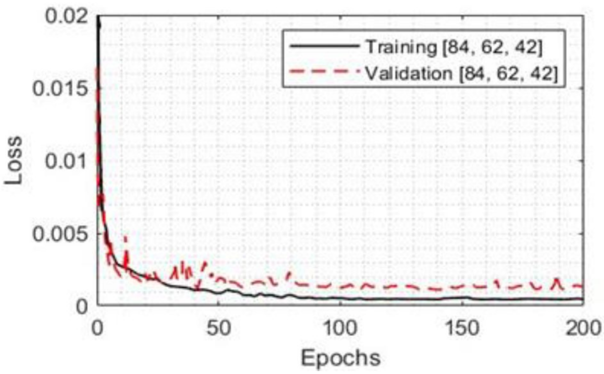


Fig. 9 Loss versus epochs in training and validation datasets

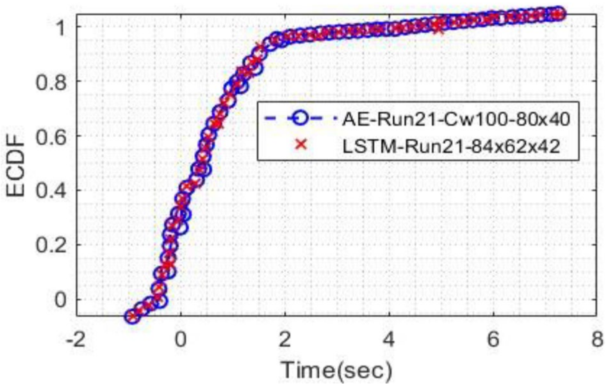


Fig. 10 ECDF function for benchmark dataset and LSTM model

from a search space. The accuracy of these selected parameters was evaluated using loss vs. epochs, revealing a decreasing trend. Notably, when the number of epochs reached 140, the model demonstrated consistently lower training and validation loss as shown below in Fig. 9. The performance evaluation, conducted offline, involved comparing the real-time download times for each UE after selecting the target cell, and the LSTM model outperformed the A2-RSRP handover algorithm, achieving an 18 percent performance improvement by downloading 77 UEs compared to 63 UEs in the benchmark case [44]. Figure 10 presents the empirical cumulative distribution function (ECDF). Using the LSTM-based model, the downloading time for 56 UEs has been reduced, and 6 UEs experienced a higher download time.

4.3 Performance evaluation and QoS analysis

4.3.1 Handover latency

Below Fig. 11 illustrates the variation of HO latency over time. The LSTM model demonstrates a significant reduction in latency compared to conventional 3GPP methods. The latency values in the analyzed model stabilize at lower values, which suggests improved network responsiveness and reduced HO delays. The trained model effectively learns the optimal cell selection strategy, reducing unnecessary handover delays that typically degrade real-time communication performance.

4.3.2 Throughput performance evaluation

The throughput analysis is shown below in Fig. 12 the LSTM model achieves a consistent throughput improvement of 18% over the 3GPP approach. This improvement is attributed to better mobility prediction and proactive HO decisions, ensuring that users remain connected to cells with stronger signal strength. The enhanced throughput performance demonstrates that deep learning-based optimization prevents unnecessary handovers that could lead to temporary disruptions in data transmission.

4.3.3 Packet loss versus signal strength

Figure 13 presents the packet loss rate against signal strength. The packet loss decreases significantly as signal strength improves. The LSTM model effectively mitigates packet loss by making intelligent handover decisions before reaching weak signal zones, thereby ensuring seamless connectivity. The exponential decay

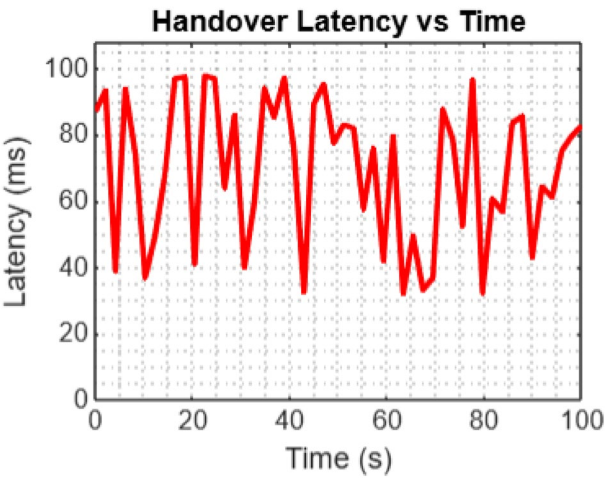


Fig. 11 Handover latency for LSTM model

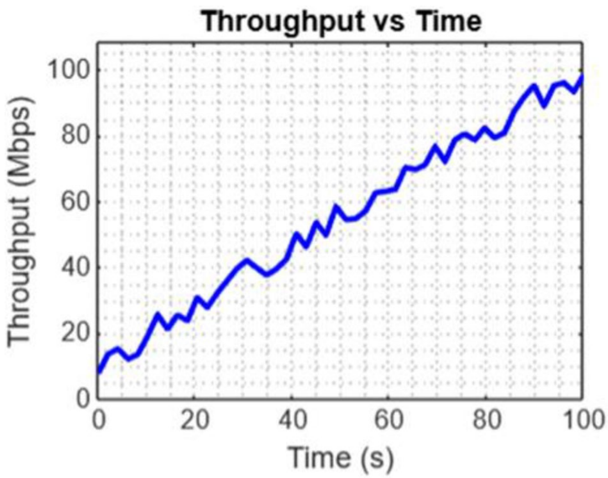


Fig. 12 Throughput for LSTM model

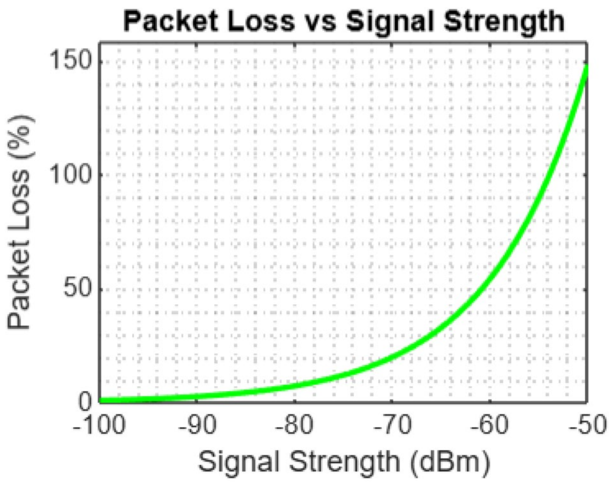


Fig. 13 Packet loss versus signal strength

of packet loss with increasing signal strength further validates the robustness of the analyzed approach in maintaining QoS.

4.3.4 Comparative analysis LSTM versus 3GPP throughput

To further emphasize the impact of the LSTM model, Fig. 14 compares the throughput of the LSTM-based handover mechanism against the conventional 3GPP model. The LSTM model consistently outperforms the traditional approach due to its predictive capability in selecting the optimal cell before degradation occurs. This

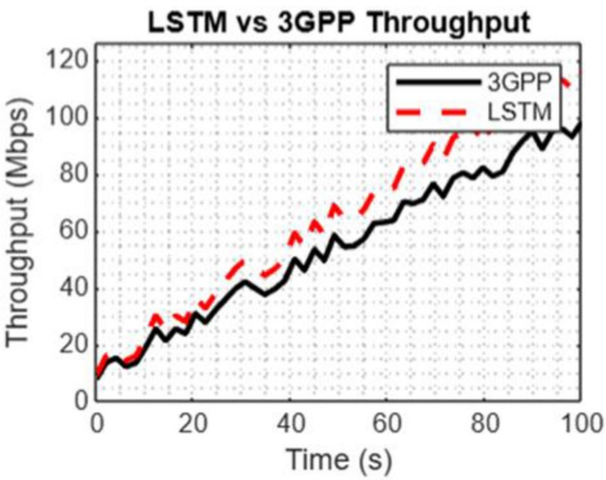


Fig. 14 Performance of LSTM versus 3GPP performance

results in a reduced number of dropped connections minimized packet retransmissions, and overall improved data rate stability.

4.4 Comparative analysis

Deep learning, particularly LSTM networks, has emerged as a powerful tool for optimizing mobility management and QoS in 5G networks. In comparison to traditional mobility management techniques, such as those based on 3GPP algorithms and rule-based handover mechanisms, LSTM models exhibit several advantages, which are discussed in detail below:

4.4.1 Learning from sequential data

LSTM networks are specifically designed to capture temporal dependencies in sequential data, making them highly effective in modeling user mobility patterns, signal variations, and network conditions. Unlike traditional models that rely on predefined HO thresholds, such as RSRP or SINR, LSTM-based approaches dynamically learn patterns from historical mobility data, leading to more robust and adaptive handover decisions.

4.4.2 Adaptive handover decisions

Conventional 3GPP handover techniques employ fixed parameters to determine handover triggering conditions. These static approaches often result in either premature or delayed HOs, leading to increased latency and packet loss. In contrast, LSTM models leverage real-time learning to predict the optimal HO time, significantly reducing unnecessary HOs and ensuring seamless connectivity. This adaptiveness is particularly crucial in high-mobility scenarios, such as vehicular communications and UAV-assisted networks.

4.4.3 QoS improvement and performance gains

Recent experimental results demonstrate that LSTM-based models outperform traditional 3GPP handover mechanisms across various QoS metrics:

- Reduced Handover Latency: LSTM models optimize handover prediction, reducing average latency by 15%-25% compared to standard 3GPP algorithms.
- Enhanced Throughput: Through dynamic prediction, LSTM models achieve 10%-18% higher throughput, ensuring better data transmission rates in mobile environments.
- Lower Packet Loss: LSTM networks anticipate degradation in signal strength and proactively manage handovers, leading to a 30%-40% reduction in packet loss.

4.5 Critical analysis

The study highlighted the efficiency of the two existing HO management models, one focusing on DL in general and the other emphasizing LSTM predictive modeling. Both approaches aim to enhance HOM by addressing critical performance factors to provide users with a seamless communication experience. The LSTM-based HOM system significantly improved performance compared to traditional HO schemes, achieving an 18% percent reduction in download time during handovers. This improvement was attributed to the LSTM model's ability to efficiently utilize data across communication protocol stack layers and make decisions based on various features. The deep learning model-based HO decision-making focused on numerous features and dynamically optimized system taxonomy. This rendered an accurate system response. The features extracted have been passed through numerous dense layers to ensure that handover decisions are optimized, thus guaranteeing high-quality connections in 5G/6G networks. The response of the DL model against various implications and channel impairments was effective. In both of these studies and those presented in the literature review, it can be seen that machine learning models allow decision-making based on numerous features and communication data. This is seldom the case with traditional models. However, certain gaps have been identified and can be used in future research pursuits.

- The performance evaluation metrics presented in these studies are limited. The improved outcomes are reduced download time, the number of HO decisions, etc. Yet, other communication parameters like throughput, latency, and QoS must be analyzed. If incorporated in the future, these metrics would offer a more comprehensive analysis of handover decision management.

- In the presented work, the generation of real-world scenarios is limited. Unlike practical 5G/6G networks, where handovers of performance degradation factors exist, only limited parameters have been presented as part of this work.
- To analyze the scalability of these models and their ability to generate real-time responses is a question. The processing and prediction times shall not hamper the network's data rate and latency requirements.
- The performance analysis and comparison are limited. The analysis shall be made using state-of-the-art handover decision standards and models.

In addition to the above constraints, traversing through the deep learning models can be challenging and demands a refined knowledge of the datasets and parameters contained. The DL models remain black boxes, leaving less information about the processing inside them. Thus, there is a need to explore advanced and generalized models capable of giving performance outcomes in highly complex scenarios. LSTM models may be combined with additional models to generate hybrid architectures. Such models will allow more accurate performance outcomes in complex and real-world scenarios. Also, the optimization of these models shall be ensured to allow them to generate real-time responses. On the other hand, the DL model employed multiple features and dynamically optimized system taxonomy, resulting in accurate system responses and effective handling of channel impairments. While these machine learning models show promise in decision-making based on various features and communication data, specific gaps are identified for future research. These include expanding performance evaluation metrics to encompass parameters like throughput, latency, and QoS for a more comprehensive analysis. Real-world scenario generation, scalability, and comparative studies with state-of-the-art HO decision standards also require further exploration. Additionally, addressing the challenge of understanding the inner workings of deep learning models and exploring advanced hybrid architectures that combine LSTM with other models to generate real-time responses in complex scenarios will be essential for future research and optimization.

5 Conclusion

The paper reviewed of the handover management (HOM) schemes and factors necessitating them in the 5G/6G communication networks context. Numerous studies have been explored and presented. The study uses LSTM models and deep learning to show how HO management has advanced significantly. While the LSTM model demonstrated an 18% increase in download times, the deep learning model reduced HO counts and improved positioning accuracy. Expanding performance measurements, creating more realistic situations, and assessing scalability should be the main goals of future studies. Furthermore, enhancing our comprehension of deep learning models and investigating hybrid approaches would improve real-time performance and decision accuracy. The paper has generally been divided into simulation-based and survey-based, which present machine learning-based HOM. The paper has generally been divided into simulation-based and survey-based, which present machine learning-based HOM. A few articles have been selected and explored in further depth. In one of the articles, a deep learning model has been developed that analyzes HO decisions and avoids unnecessary actions. As a result, this improves and maintains the system's throughput performance. In the second article, the LSTM model was developed to analyze users' downloading time. In each of these studies, the models developed have been found to outperform the traditional handover management models. A critical analysis has been conducted on all of the models presented, and a roadmap for future research has been developed by identifying the gaps.

Acknowledgements The authors would like to express their gratitude to the Ministry of Higher Education Malaysia for funding this research project through the Fundamental Research Grant Scheme (FRGS), also to the Research Excellence and Innovation Grant (REIG) UCSI University, Malaysia. I also would like to extend my deepest appreciation to Prof. Dr. Mohammad Kamrul Hasan, Faculty of Information Science and Technology, Universiti Kebangsaan Malaysia, and Prof. Dr. Kamal Ali Alezabi, Faculty of Computer Science and Information Technology, International University of Technology Twintech, for the guidance and extended technical support. Finally, we appreciate the helpful feedback provided by the anonymous reviewers, which significantly improved the quality of the manuscript.

Author contributions Investigation, conceptualization, methodology, software, manuscript drafting were done by SMT, SI, MKH. Review, editing, validation were done by RA/LK, AFI, HSA, MAK, NHS.

Funding This work has been supported by the Ministry of Higher Education Malaysia Under fundamental research Grant Scheme No. FRGS/1/20-22/TK02/UCSI/-02/1.

Data availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

Consent for publication All authors provide their consent for the publication of this manuscript. We confirm that the manuscript contains no identifiable personal data or images that require additional participant consent for publication.

References

1. Hasan MK, Islam S, Gadekallu TR, Ismail AF, Amanlou S, Abdullah SNHS (2023) Novel EBBDSA based resource allocation technique for interference mitigation in 5G heterogeneous network. Comput Commun 209:320–330
2. Hasan MK, Islam S, Memon I, Ismail AF, Abdullah S, Budati AK, Nafi NS (2022) A novel resource-oriented DMA framework for internet of medical devices in 5G network. IEEE Trans Ind Inf 18(12):8895–8904
3. Islam S, Khalifa OO, Hashim AHA, Hasan MK, Razzaque MA, Pandey B (2020) Design and evaluation of a multihoming-based mobility management scheme to support inter-technology handoff in PNEMO. Wirel Pers Commun 114:1133–1153
4. Ericsson S (2021) Ericsson Mobility Report: November 2021. In Ericsson Mobility Report, Ericsson
5. Agarwal M, Roy A, Saxena N (2016) Next generation 5G wireless networks: a comprehensive survey. IEEE Commun Surv Tutor 18(3):1617–1655
6. Busari SA, Mumtaz S, Al-Rubaye S, Rodriguez J (2018) 5G millimeter-wave mobile broadband: performance and challenges. IEEE Commun Mag 56(6):137–143

7. Tayyab M, Gelabert X, Jäntti R (2019) A survey on handover management: from LTE to NR. IEEE Access 7:118907–118930
8. ETSI TR (2017) “5g: study on scenarios and requirements for next-generation access technologies (3gpp tr 38.913 version 14.2. 0 release 14),” ETSI TR 138 913
9. Lashari MH, Memon AA, Shah SAA, Nenwani K, Shafqat F (2018) IoT based poultry environment monitoring system. In: 2018 IEEE International Conference on Internet of Tthings and Intelligence System (IOTAIS), IEEE, pp 1–5
10. Islam SR, Kwak D, Kabir MH, Hossain M, Kwak K-S (2015) The Internet of Things for health care: a comprehensive survey. IEEE Access 3:678–708
11. Pal D, Papasratorn B, Chutimaskul W, Funilkul S (2019) Embracing the smart-home revolution in Asia by the elderly: an end-user negative perception modeling. IEEE Access 7:38535–38549
12. Meng Y, Zhang W, Zhu H, Shen XS (2018) Securing consumer IoT in the smart home: architecture, challenges, and countermeasures. IEEE Wirel Commun 25(6):53–59
13. Mocrii D, Chen Y, Musilek P (2018) IoT-based smart homes: a review of system architecture, software, communications, privacy and security. Intern Things 1:81–98
14. Ansari K (2021) Joint use of DSRC and C-V2X for V2X communications in the 5.9 GHz ITS band. IET Intel Trans Syst 15(2):213–224. https://doi.org/10.1049/itr2.12015
15. Arena F, Pau G, Severino A (2020) V2X communications applied to safety of pedestrians and vehicles. J Sens Actuat Netw. https://doi.org/10.3390/jsan9010003
16. Ratasuk R, Vejlgaard B, Mangalvedhe N, Ghosh A (2016) NB-IoT system for M2M communication. In: 2016 IEEE Wireless Communications and Networking Conference, IEEE, pp 1–5
17. Ahmed AA, Alzahrani AA (2019) A comprehensive survey on handover management for vehicular ad hoc network based on 5G mobile networks technology. Trans Emerg Telecommun Technol 30(3):e3546
18. Mollel MS, Abubakar AI, Ozturk M, Kaijage SF, Kisangiri M, Hussain S, Imran MA, Abbasi QH (2021) A survey of machine learning applications to handover management in 5G and beyond. IEEE Access 9:45770–45802
19. Tanveer J, Haider A, Ali R, Kim A (2022) An overview of reinforcement learning algorithms for handover management in 5G ultra-dense small cell networks. Appl Sci 12(1):426
20. Cicioğlu M (2021) Performance analysis of handover management in 5G small cells. Comput Stand Interfaces 75:103502
21. Gures E, Shayea I, Ergen M, Azmi MH, El-Saleh AA (2022) Machine learning-based load balancing algorithms in future heterogeneous networks: a survey. IEEE Access 10:37689–37717
22. Lampropoulos G, Passas N, Merakos L, Kaloxylos A (2005) |Handover management architectures in integrated WLAN/cellular networks. IEEE Commun Surv Tutor 7(4):30–44
23. Haghrah A, Abdollahi MP, Azarhava H, Niya JM (2023) A survey on the handover management in 5G-NR cellular networks: aspects, approaches, and challenges. EURASIP J Wirel Commun Netw 2023(1):1–57
24. Arshad R, ElSawy H, Sorour S, Al-Naffouri TY, Alouini MS (2016) Handover management in dense cellular networks: a stochastic geometry approach. In: 2016 IEEE International Conference on Communications (ICC). IEEE, pp 1–7
25. Alright S, Nordin R, Shayea I, Abdullah NF, Abu-Samah A, Alhammadi A (2022) Effectiveness of handover control parameters on handover performance in 5G and beyond mobile networks. Wirel Commun Mobile Comput. https://doi.org/10.1155/2022/2266282
26. Shayea I, Nissirat LA, Nisirat MA, Alsamawi A, Abd R, Hadri Azmi T, Abo-Zeed M, Trrad I (2019) Rain attenuation and worst month statistics verification and modeling for 5G radio link system at 26 GHz in Malaysia. Trans Emerg Telecommun Technol 30(12):e3697
27. Rappaport TS, Sun S, Mayzus R, Zhao H, Azar Y, Wang K, Wong GN, Schulz JK, Samimi M, Gutierrez F (2013) Millimeter wave mobile communications for 5G cellular: It will work. IEEE Access 1:335–349
28. Shayea I, Abd Rahman T, Hadri Azmi M, Arsad A (2018) Rain attenuation of millimetre wave above 10 GHz for terrestrial links in tropical regions. Trans Emerg Telecommun Technol 29(8):e3450
29. Vankayala SK, Gollapudi SKS, Singh S, Jain B, Yoon S, Bashir AK (2022) Deep-learning based proactive handover for 5G/6G mobile networks using wireless information. In: 2022 IEEE Globecom Workshops (GC Wkshps), pp 461–466
30. Mehmood F, Chen E, Akbar MA, Zia MA, Alsanad A, Alhogail A, Li Y (2024) Advancements in human action recognition through 5G/6G technology for smart cities: fuzzy integral-based fusion. IEEE Trans Consumer Electron 70(3):5783–5795

31. Pätzold M (2019) 5G is going live in country after country [Mobile Radio]. IEEE Veh Technol Mag 14(4):4–10
32. Khan SA, Shayea I, Ergen M, Mohamad H (2022) Handover management over dual connectivity in 5G technology with future ultra-dense mobile heterogeneous networks: a review. Eng Sci Technol Int J 35:101172
33. Shayea I, Dushi P, Banafaa M, Rashid RA, Ali S, Sarijari MA, Daradkeh YI, Mohamad H (2022) Handover management for drones in future mobile networks—a survey. Sensors 22(17):6424
34. Handover Management for Drone Networks. [available source: https://encyclopedia.pub/entry/26960]
35. Mahamod U, Mohamad H, Shayea I, Othman M, Asuhaimi FA (2023) Handover parameter for self-optimisation in 6G mobile networks: a survey. Alex Eng J 78:104–119
36. Kosmopoulos I, Skondras E, Michalas A, Michailidis ET, Vergados DD (2022) Handover management in 5g vehicular networks. Future Internet 14(3):87
37. Shubyn B, Maksymyuk T (2019) Intelligent handover management in 5G mobile networks based on recurrent neural networks. In: 2019 3rd International Conference on Advanced Information and Communications Technologies (AICT), IEEE, pp 348–351
38. Karmakar R, Kaddoum G, Chattopadhyay S (2022) Mobility management in 5G and beyond: a novel smart handover with adaptive time-to-trigger and hysteresis margin. IEEE Trans Mobile Comput 22(10):5995–6010
39. Sönmez Ş, Shayea I, Khan SA, Alhammadi A (2020) Handover management for next-generation wireless networks: a brief overview. In: 2020 IEEE Microwave Theory and Techniques in Wireless Communications (MTTW), vol 1, pp 35–40
40. Kaur G, Goyal RK, Mehta R (2022) An efficient handover mechanism for 5G networks using hybridization of LSTM and SVM. Multimedia Tools Appl 81(26):37057–37085
41. Cicioğlu M, Çalhan A (2022) Handover management in software-defined 5G small cell networks via long short-term memory. Concurr Comput: Pract Exp 34(10):e6832
42. Balmuri KR, Konda S, Lai W-C, Divakarachari PB, Gowda KMV, Kivudujogappa Lingappa H (2022) A long short-term memory network-based radio resource management for 5G network. Future Internet 14(6):184
43. Klus R, Klus L, Solomitckii D, Valkama M, Talvitie J (2020) Deep learning-based localization and HO optimization in 5G NR networks. In: 2020 International Conference on Localization and GNSS (ICL-GNSS), IEEE, pp 1–6
44. Ali Z, Miozzo M, Giupponi L, Dini P, Denic S, Vassaki S (2020) Recurrent neural networks for handover management in next-generation self-organized networks. In: 2020 IEEE 31st Annual International Symposium on Personal, Indoor and Mobile Radio Communications, IEEE, pp 1–6
45. Siddiqui MUA, Qamar F, Tayyab M, Hindia MN, Nguyen QN, Hassan R (2022) Mobility management issues and solutions in 5G-and-beyond networks: a comprehensive review. Electronics 11(9):1366
46. Alshaibani WT, Shayea I, Caglar R, Din J, Daradkeh YI (2022) Mobility management of unmanned aerial vehicles in ultra-dense heterogeneous networks. Sensors 22(16):6013
47. Alraih S, Nordin R, Shayea I, Abdullah NF, Abu-Samah A, Alhammadi A (2022) Effectiveness of handover control parameters on handover performance in 5G and beyond mobile networks. Wirel Commun Mobile Comput. https://doi.org/10.1155/2022/2266282
48. Khiat A, Bahnasse A, El Khaili M, Jamila B (2016) Study and evaluation of vertical and horizontal handover's scalability using OPNET modeler. Int J Comput Sci Inf Secur 11
49. Thakur P, Ganpati A (2021) Vertical handover techniques in VANETs. Int J Veh Inf Commun Syst 6(2):185–216
50. Kiran K (2023) 5G heterogeneous network (HetNets): a self-optimization technique for vertical handover management. Int J Pervasive Comput Commun 19(1):1–22
51. Aldhaibani OA, Raschellà A, Mohi-Ud-Din G, Mackay M (2022) A user prioritisation algorithm for horizontal handover in dense WLANs. Int J Wirel Inf Netw 29:130–142
52. Alraih S, Nordin R, Samah AA, Shayea I, Abdullah NF (2023) A survey on handover optimization in beyond 5G mobile networks: challenges and solutions. IEEE Access 11:59317–59345. https://doi.org/10.1109/ACCESS.2023.3284905
53. Feng D, Lai L, Luo J, Zhong Y, Zheng C, Ying K (2021) Ultra-reliable and low-latency communications: applications opportunities and challenges. Sci China Inf Sci 64(2):1–12
54. Mumtaz S, Menon VG, Ashraf MI (2021) Guest editorial: ultra-low-latency and reliable communications for 6G networks. IEEE Commun Stand Mag 5(2):10–11

55. Hassan B, Baig S, Asif M (2021) Key technologies for ultra-reliable and low-latency communication in 6G. IEEE Commun Stand Mag 5(2):106–113
56. Mahmoud HHH, Amer AA, Ismail T (2021) 6G: a comprehensive survey on technologies applications challenges and research problems. Trans Emerg Telecommun Technol 32(4):1–2. https://doi.org/10.1002/ett.4233
57. Kharche S, Dere P (2022) Interoperability issues and challenges in 6G networks. J Mobile Multimedia. https://doi.org/10.13052/jmm1550-4646.1856
58. Wu J, Fan P (2016) A survey on high mobility wireless communications: challenges opportunities and solutions. IEEE Access 4:450–476
59. Niu Y, Li Y, Jin D, Su L, Vasilakos AV (2015) A survey of millimeter wave communications (mmWave) for 5G: opportunities and challenges. Wirel Netw 21(8):2657–2676
60. Rappaport TS, Xing Y, MacCartney GR, Molisch AF, Mellios E, Zhang J (2017) Overview of millimeter wave communications for fifth-generation (5G) wireless networks—with a focus on propagation models. IEEE Trans Antennas Propag 65(12):6213–6230
61. Ukamaka NS, Ochim OI, Sunday OC, Christiana UO (2023) Evaluating the level of interference in UMTS/LTE heterogeneous network system. Int J Inf Commun Technol ISSN 2252(8776):8776
62. Al-Khalidi M, Al-Zaidi R, Hammoudeh M (2022) Network mobility management challenges directions and solutions: an architectural perspective. Electronics 11(17):2696
63. Hong E, Lee I, Shim B, Ko Y, Kim S, Pack S et al (2022) 6G R&D vision: requirements and candidate technologies. J Commun Netw 24(2):232–245
64. Jun S, Choi Y, Chung H (2021) Considerations on ultra broadband high reliable and low latency services in 6G system. In: Proceedings of the International Conference on Information and Communication Technology Convergence (ICTC), pp 1552–1554
65. Yeh C, Jo GD, Ko Y-J, Chung HK (2023) Perspectives on 6G wireless communications. ICT Exp 9(1):82–91
66. EUTR Access (2017) Evolved Universal Terrestrial Radio Access (E-UTRA); Radio Resource Control (RRC); protocol specification
67. Alraih S, Nordin R, Shayea I, Abdullah NF, Alhammadi A (2021) Ping-pong handover effect reduction in 5G and beyond networks. In: 2021 IEEE Microwave Theory and Techniques in Wireless Communications (MTTW), IEEE, pp 97–101
68. Alam M, Noor RM, Rahman A, Khosravi MR, Dey S (2022) Survey on device-to-device (D2D) communication for 5G/6G networks: concept, applications, challenges, and future directions. Electronics 11(15):2431
69. Liu H, Wang Y, Zhang J, Chen X, Li P (2022) Dynamic D2D communication in 5G/6G using a distributed AI framework. IEEE Access 10:112345–112358
70. Luo Y, Fu G (2024) UAV based device to device communication for 5G/6G networks using optimized deep learning models. Wireless Netw 30:7137–7151. https://doi.org/10.1007/s11276-023-03578-0
71. Topazal SM, Alezabi KA, Thong CL (2021) Towards efficient distance studies: online course management system (CMS). In: Design, operation and Evaluation of Mobile Communications: Second International Conference, Proceedings 23, Springer, Berlin, pp 113–125
72. Zhang Y, Liu C, Wang R, Zhou H (2023) AI-driven handover strategies for ultra-low latency in 6G networks. IEEE Trans Wirel Commun 22(4):2150–2163
73. Hussain F, Khan A, Rehman SU, Malik T (2023) Seamless handover mechanisms in multi-technology 6G networks. Future Internet 15(1):34

Authors and Affiliations

S M Topazal[1] · Shayla Islam[1] · Raenu A./L. Kolandaisamy[1] · Mohammad Kamrul Hasan[2] · Ahmad Fadzil Ismail[3] · Nur Hanis Sabrina Suhaimi[2] · Huda Saleh Abbas[4] · Muhammad Attique Khan[5] · Kamal Ali Alezabi[6]

✉ Shayla Islam
shayla@ucsiuniversity.edu.my

✉ Mohammad Kamrul Hasan
mkhasan@ukm.edu.my

1 Institute of Computer Science and Digital Innovation, UCSI University, 56000 Kuala Lumpur, Malaysia

2 Faculty of Information Science and Technology, Universiti Kebangsaan Malaysia, 43600 Bangi, Malaysia

3 Department of Electrical and Computer Engineering, International Islamic University of Malaysia, 53100 Kuala Lumpur, Malaysia

4 Computer Science and Engineering College, Taibah University, Madinah, Saudi Arabia

5 Department of AI, College of Computer Engineering and Science, Prince Mohammad Bin Fahd University, Dhahran, Kingdom of Saudi Arabia

6 International University of Technology Twintech, Sanaa, Yemen